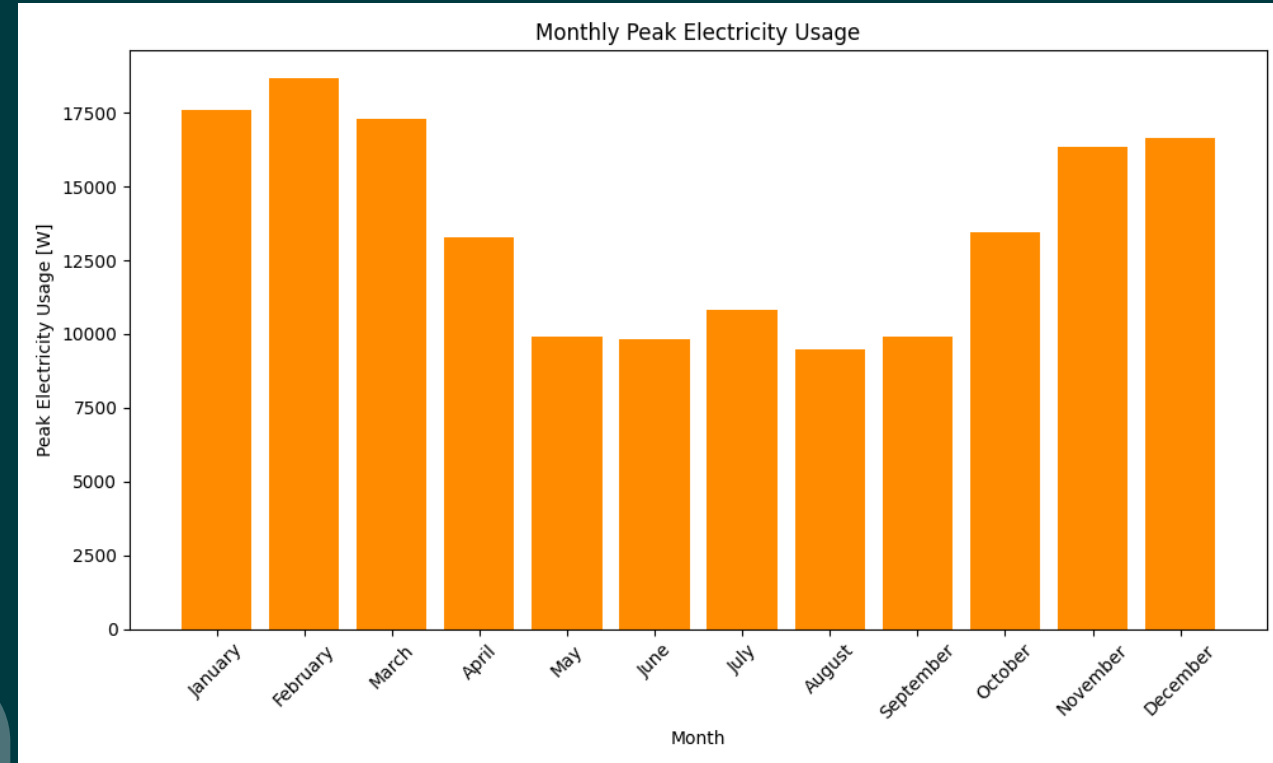


SIMPARC FOR ALBERTA

BACKGROUND

- Residential housing stock models are an integral part to understanding energy policy within a region
- They help developers understand
 - Predicted baseline energy
 - Forecasted growth rates
- This helps with
 - Managing housing supply
 - Identifying peak energy periods
 - Sustainability initiatives
- The HydroQuebec SimParc model is a residential housing stock model based in Quebec
 - Build on Resstock framework
 - Contains CSV generator and SimParc model
 - Used to predict energy data in Quebec



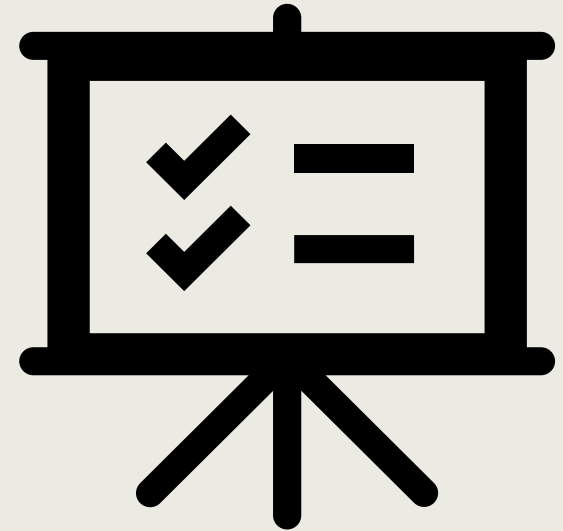
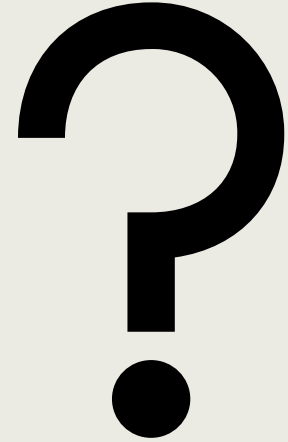
*Peak Monthly Electricity usage for an Apartment Building in Montreal
Year Built: 1955
Conditioned floors: 1
Conditioned floor area: 105.72 sqm*

PROBLEM

- Alberta needs a housing stock energy model curated to its specific climate and residential usage patterns
- The HydroQuebec SimParc model is inaccessible due to:
 - French context
 - Scaling concerns: Streamlit bottlenecks, high memory usage
 - Hardcoded Quebec data – EUEMR Survey

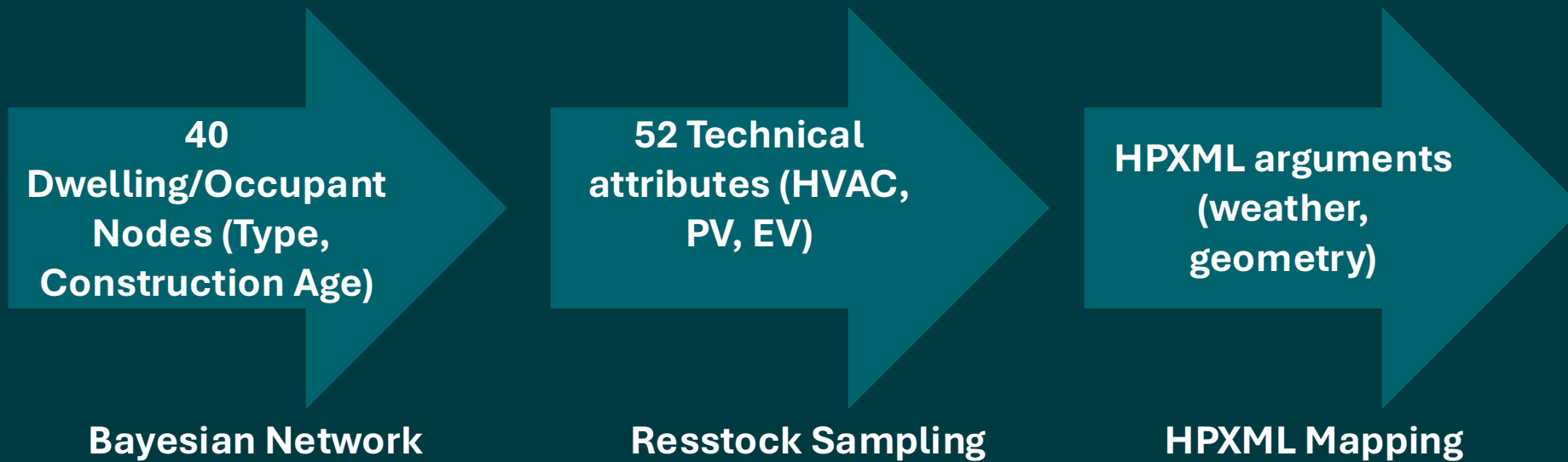
OBJECTIVE

- Adapt the Resstock based SimParc model to utilize residential energy data from the City of Calgary for a baseline simulation



METHODS - DISSECTION

1. The Sampler is a front end for SimParc
 - Main model in CSV Generator
2. Draws 219 HPXML arguments per home



3. Bayesian Network Based on EUEMR survey
 - New data needed

METHODS – DATA COLLECTION

Problem: No Albera EUEMR



Substitute: NRCAN Energuide data, 2021 Census



Housing feature	EnerGuide sample	Calgary census
Single-detached	82.7%	55.2%
Apartment (Collective)	0.2%	27.1%
Row house	11.6%	8.8%
Built $\geq$ 2020	21.1%	2.4%

Solution: Re-weight energuide data to match census



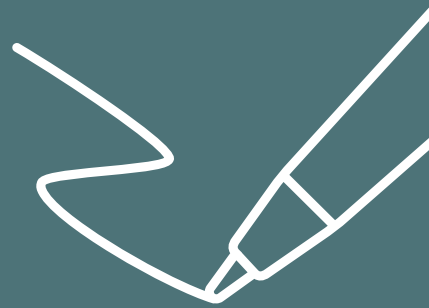
METHODS – DATA REWEIGHT

METHODS – NEXT STEPS

Identify equivalent Albertan datasets with the same or higher spatial and temporal resolution



Rewrite doMapping with Albertan parameters



Run Bayesian network on City of Calgary data

